Arista Access Node WFD-ACC-59 Whitefield POP Optical Transport Failure SOP

Node: WFD-ACC-59 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_1370

Description

Optical transport failures on Arista access nodes, specifically in the context of WFD-ACC-59 at the Whitefield POP, can arise from a variety of factors related to the SFP/XFP modules or the underlying fiber infrastructure. These failures can be attributed to degradation of the SFP/XFP modules, fiber cuts, or loss of receive and transmit power. The SFP/XFP modules play a crucial role in optical transport, as they are responsible for converting electrical signals into optical signals that can be transmitted over fiber optic cables. The receive and transmit power levels are essential for maintaining signal integrity and preventing signal degradation. In the event of an optical transport failure, it is necessary to diagnose the root cause and perform corrective actions to restore normal operations. The following SOP outlines the procedures for diagnosing and resolving optical transport failures on Arista access nodes.

Pre-requisite Checks

- Check: Verify the Whitefield POP's physical environment for any signs of damage or anomalies that could be contributing to the optical transport failure.

- Check: Confirm that the redundant link is operational and functioning correctly.

- Check: Gather information from the network monitoring tools regarding the timing and duration of the optical transport failure.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP modules	High
Fiber cuts or damage	Medium
Loss of receive and transmit power	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface SFP, Expected Output: SFP information, including status and error counters., Description: This command will provide information about the SFP module, including its status and any error counters that may indicate a problem.

Command: show interface fiber, Expected Output: Fiber optic link information, including transmit and receive power levels., Description: This command will provide information about the fiber optic link, including the transmit and receive power levels, which can help diagnose issues related to signal degradation.
```

```
Command: show log, Expected Output: System log information, including error messages related to the optical transport failure., Description: This command will provide system log information, including any error messages that may be related to the optical transport failure.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: no interface SFP replace, Description: This command will remove the faulty SFP module and replace it with a new one.

Step 2: Step: Verify fiber integrity, Command: show interface fiber, Description: This command will verify the integrity of the fiber optic link and ensure that the transmit and receive power levels are within the recommended range.

Step 3: Step: Reset the interface, Command: interface no shutdown, Description: This command will reset the interface and bring it back online.

Post-Check Verification

```
Command: show interface , Expected Output: Interface information, including status and error counters., Description: This command will verify that the interface is functioning correctly and that there are no error counters indicating a problem.

Command: show log, Expected Output: System log information, including error messages related to the optical transport failure., Description: This command will verify that the system log does not contain any error messages related to the optical transport failure.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: interface no replace, Description: This command will restore the original SFP module that was replaced during the resolution step.

- Step: Reset the interface, Command: interface no shutdown, Description: This command will reset the interface to its original state.